

Stockwise: Beginner's Guide to Investing in Stocks



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Abstract

Stockwise is an investment app designed to support beginner investors by providing financial information that is more accessible and easier to understand. Many beginner investors often struggle to understand investment risk, stock analysis, and market trends because they lack experience and financial knowledge. They may also avoid professional investment advice because it can be expensive. To help solve this problem, Stockwise provides simplified financial data, educational resources, market news, search tools, and intelligent support features, such as AI, in one platform. The app is designed to help users learn about investing, analyze stocks, gain useful market insights, and make better-informed investment decisions. The Stockwise application aims to improve financial understanding and increase confidence among beginner investors by presenting complex financial information in a clear, user-friendly way

1.0 Introduction

Financial freedom is what most people strive to achieve in their lifetimes. Having less debt and financial stability can give people greater freedom, a better lifestyle, and more opportunities to enjoy their time. One way to achieve this goal is through investing, as a good investment can help build wealth over time. However, investing is not easy for beginners. Many beginner investors often lack experience, knowledge, and a proper understanding of risk, leading them to make irrational decisions that could devastate their personal finances. Research shows that rational investment decisions are not made through emotion. Irrational decisions are influenced by behavioral biases such as overconfidence, herding, and representativeness. These mistakes can lead to poor investment choices and may seriously harm personal finances.

1.1 Background

Investing can help people grow their money and achieve long-term financial goals. Nowadays, beginner investors do not fully understand investment risks, market trends, or how to properly analyze stocks. Without this understanding, the risk of financial loss is very high. However, beginner investors do not often seek an investment advisor because professional financial guidance can be expensive. Therefore, an investment app can serve as a practical tool to help beginners learn about the market, gain investment insights, and make better-informed investment decisions.

1.2 Motivation

Stockwise is designed to assist beginner investors by offering simplified financial data, educational resources, and research tools that make investing easier to understand and more accessible. Many new beginner investors often lack the experience needed to analyze stocks, understand market conditions, or interpret financial information correctly. This platform allows users to examine stocks, view important market data, and stay updated on financial news in one place. By organizing complex financial information into a clear and user-friendly format, Stockwise helps beginners learn about investing, gain useful market insights, make more informed decisions, and build confidence in managing their investments.

1.3 Related Work

1.3.1 Utilizing Artificial Intelligence to Provide Financial Insights and Stock

Artificial Intelligence has been in the headlines nowadays. One of the research papers has focused on using artificial intelligence to provide financial insights and stock predictions. The research paper, Generative AI Chatbot Using Retrieval Augmented Generation for Financial Insights, studied how a financial chatbot can use Retrieval-Augmented Generation, large language models, and financial datasets to answer user questions. Similarly, the paper Significance of Predictors: Revisiting Stock Return Predictions Using Explainable AI uses Explainable AI, deep learning, and ensemble methods to identify the most important factors in stock return prediction and improve prediction accuracy. These techniques can be implemented in Stockwise

because artificial intelligence can help beginner investors understand about financial information, gain market insights, and make better-informed investment decisions.

1.3.2 Using Machine Learning for Stock Trend and Risk Prediction

Other research has focused on using machine learning to predict stock trends and investment risk. The research paper, *Analysing the Trend of Stock Market and Evaluate the Performance of Market Prediction using Machine Learning Approach* studied how to use machine learning methods such as Random Forest, Support Vector Machine, and Neural Network can be used to predict future stock values. Similarly, the *Risk Prediction of Investment System under Big Data Statistical Algorithms* paper uses statistical models and risk evaluation methods to assess investment risk and support better decision-making. These techniques can be applied in Stockwise because data-driven methods can help beginner investors understand stock performance, analyze market trends, and evaluate investment risk.

2.0 Problem Statement

Many beginner investors want to grow their money quickly, but they often lack the knowledge and experience needed to make informed investment decisions and achieve long-term financial goals. Understanding investment risks, market trends, and stock analysis can be difficult for beginners, especially when financial information is complex and spread across many sources. In addition, professional financial advice can be expensive, which makes it less accessible to many new investors. As a result, beginner investors may rely on incomplete information or make poor decisions that increase the risk of financial loss. Hence, since computing costs are much lower nowadays, there is a need for an investment app that provides easy access to new investors and offers simplified market information, educational resources, and helpful insights to support better decision-making

3.0 Proposed Project & Significance

The purpose of Stockwise is to bridge the gap between complex financial markets and beginner investors by providing a simple, intuitive platform for learning and decision-making. Many new investors are overwhelmed by the risk of making uninformed choices. Stockwise addresses this by centralizing key features such as stock analysis, real-time data, market news, and personalized recommendations into one accessible mobile application. Its goal is not only to provide information, but to present it in a way that is easy to understand.

The significance of this project is to improve financial literacy and make the idea of investing not as difficult as it seems, so individuals can take control of their financial futures. By offering educational resources alongside practical tools, Stockwise helps users build foundational knowledge while actively engaging with the market. This reduces reliance on guesswork or unreliable sources and encourages more informed, confident investment decisions. Additionally, the integration of user preferences and recommendation systems enhances the user experience by tailoring content to individuals' interests, making the platform more relevant and effective.

Overall, Stockwise is significant because it promotes smarter investing habits, lowers the barrier to entry for the stock market, and supports users in developing long-term financial skills that can contribute to personal financial growth and stability.

4.0 Objectives (Step-by-step requirements and UX)

Overall Project Objective

The objective of Stockwise is to develop a user-friendly mobile application that helps beginner investors understand and navigate the stock market with confidence. The project aims to integrate real-time market data, news, and personalized recommendations into a single platform, allowing users to research stocks, track trends, and make informed investment decisions. It focuses on simplifying complex financial concepts through clear visuals and educational resources through our Stockwise AI, ultimately promoting smarter investing habits and improving financial literacy among new users. Cutting out the middleman, who receives hundreds to thousands of dollars, by just reading the stock market trends, which you can now do on your own.

4.1 UX Design and functions

4.1.1 Getting Started with Stockwise

Users begin by creating an account or logging into the application. The process is simple and guides users through setting preferences such as investment interests and experience level. Once logged in, users can access a dashboard that displays key stock information, trending assets, and relevant financial news.



Figure (4.1.1.1) User account creation

4.1.2 Personalized User Profile

Users create a personalized profile by selecting their interests, preferred industries, and investment goals. Based on this data, Stockwise provides the user with tailored recommendations, curated news, and relevant educational content. It also blocks irrelevant news articles.

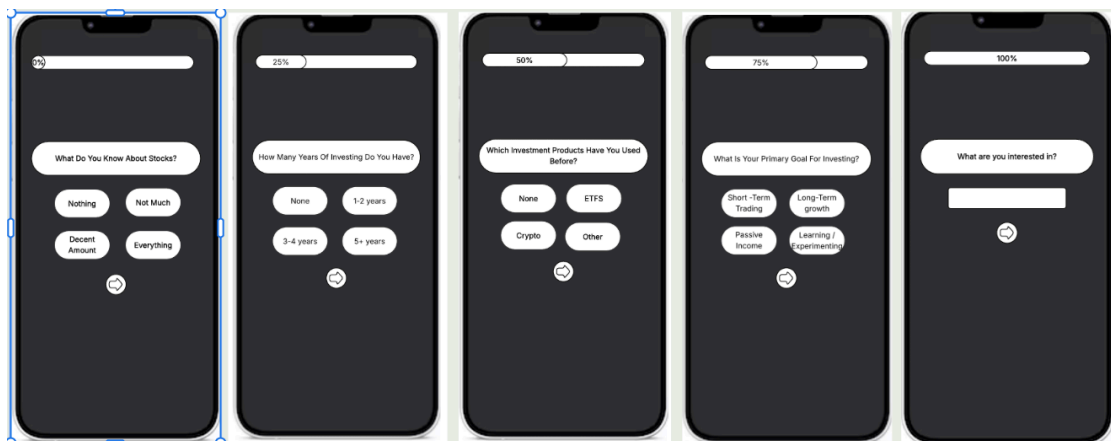


Figure (4.1.1.2) User queries

4.1.3 User Queries and Stock Search

Users can search for specific stocks using a search bar that is tailored to their interests. The application provides detailed stock data, including price trends, performance charts, and relevant news articles. This feature allows users to quickly analyze and compare stocks. It is not tailored, just for local articles; it also has access to other media around the world, which is translated to the user's preferred language of choice. Because of this, the user has access to more data.

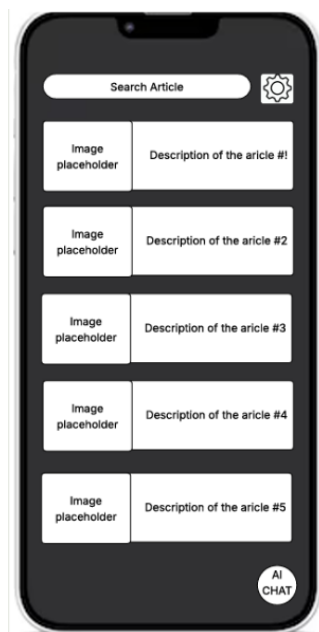


Figure (4.1.1.3) article search

4.1.4 Dashboard and Visualization

The main dashboard is designed to simplify how users understand financial information by presenting data through clear graphs and charts instead of raw numbers or complex tables. This visual approach helps beginners quickly interpret stock performance, identify patterns, and compare changes over time without needing advanced financial knowledge. There are also features such as the return on interest (ROI) calculator, which shows how much the stock has gone up or down since the stock was made public. If the stock is a “newer” one, then it would flag it as a higher risk-to-reward ratio, due to the fact that the stock can plummet or rise in value based on current states of the world. There is also a slider to show how much the user is expected to get based on how the stock has increased in the past.



Figure(4.1.1.4) stock market information

4.1.5 StockwiseAI

Stockwise AI is integrated as a simple chat-based assistant within the app to provide users with quick and intuitive access to information without needing to navigate away from the platform. Its interface is designed to be clean and easy to use, allowing users to type questions and receive instant responses, which makes it especially helpful for beginner investors. It serves as both an educational and support tool by explaining financial terms, clarifying stock data, and guiding users on how to use different features of the app. No other inquiries are allowed because it is a trained AI model that will answer all things finance-related.



Figure(4.1.1.5) Stockwise AI

5.0 Activities

5.1. Functionality

Stockwise provides core functionalities such as user authentication, stock search, real-time data retrieval, news integration, and personalized recommendations. These features work together to create a seamless and informative user experience.

5.2. User-friendly

The application emphasizes simplicity and ease of use. The interface is designed with clear navigation, minimal clutter, and visual aids such as charts and graphs to help users understand financial data without confusion.

5.3. Security

Stockwise ensures user data protection through secure authentication systems and encrypted data storage. User information, such as account details and preferences are handled securely to maintain privacy and trust.

6.0 References

- A. K. S, P. Krishna and N. S, "Generative AI Chatbot Using Retrieval Augmented Generation for Financial Insights," 2025 IEEE 9th International Conference on Information and Communication Technology (CICT), Chennai, India, 2025, pp. 1-6, doi:10.1109/CICT67193.2025.11398968
- Goswami, Bhaskar, and Ajim Uddin. "Significance of Predictors: Revisiting Stock Return Predictions Using Explainable AI." *Annals of Operations Research* [New York], vol. 357, no. 1, February 2026, pp. 223–57, <https://doi.org/10.1007/s10479-025-06717-2>
- L. Mathanprasad and M. Gunasekaran, "Analysing the Trend of Stock Market and Evaluate the performance of Market Prediction using Machine Learning Approach," 2022 International Conference on Advances in Computing, Communication and Applied Informatics (ACCAI), Chennai, India, 2022, pp. 1-9, doi:10.1109/ACCAI53970.2022.9752616
- Sugangga, R., Rahman, I., & Makaryanawati, M. (n.d.). *Literature study of stock investment risk analysis for beginner investors*. JMK (Jurnal Manajemen dan Kewirausahaan). <https://doi.org/10.32503/jmk.v9i1.5352>
- Y. Zhao, "Risk Prediction of Investment System Under Big Data Statistical Algorithms," 2024 3rd International Conference on Artificial Intelligence and Autonomous Robot Systems (AIARS), Bristol, United Kingdom, 2024, pp. 308-313, doi:10.1109/AIARS63200.2024.00064